

ABHAV ARORA

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Education

Indian Institute of Technology , Patna

Bachelor of Technology in Artificial Intelligence and Data Science

CGPA - 8.34

2024 - 2028

Academic Global School , Gorakhpur

Senior Secondary Education

Percentage - 93.6

2024

Little Flower School , Gorakhpur

Secondary Education

Percentage - 96.2

2022

Projects

FocusNest | React, Python, On-Device LLM, DroidRun Framework

Feb. 2026

- Engineered a digital wellness agent using the **DroidRun Framework**, achieving a **45% reduction** in habitual app usage through real-time, context-aware AI interventions.
- Optimized **Python-based** model inference for mobile hardware, reducing local LLM latency by **120ms** while maintaining 98% battery efficiency on Snapdragon NPU.
- Developed a high-performance monitoring dashboard in **React**, facilitating real-time visualization of user behavior patterns and securing **Third Place** at DroidRun DevSprint.

Video-Based RAG System | Python, LangChain, ChromaDB, OpenAI API

Jan. 2026

- Architected a **Retrieval-Augmented Generation (RAG)** pipeline using **ChromaDB** to index 100+ hours of video content, enabling sub-second semantic search queries.
- Implemented persistent vector storage and collection-based sharding in **ChromaDB**, improving data retrieval throughput by **40%** for large-scale video datasets.
- Engineered a custom metadata extraction engine in **Python** that maps frame-level embeddings to timestamps, resulting in a **35% increase** in temporal search accuracy.

Full-Stack Personal Portfolio | Framer Motion, GSAP, Next.js, React

March 2026

- Designed an immersive user experience utilizing **Framer Motion** and **GSAP** for complex physics-based animations, resulting in a **50% increase** in average user session duration.
- Optimized high-frequency animation cycles using **Anime.js** and the **React Dev Library**, reducing main-thread execution time by **25ms** during complex scene transitions.
- Architected a high-performance frontend achieving a **98/100 Lighthouse score** by leveraging advanced memoization and asset-loading strategies for heavy animation libraries.

Technical Skills

Languages: C++, Python, JavaScript (ES6+), Java, SQL (MySQL)

AI & Machine Learning: RAG Pipelines, Vector Databases (ChromaDB , Pinecone), LangChain, PyTorch

Web Development: React, Next.js, Node.js, Tailwind CSS, RESTful APIs, Headless CMS

Developer Tools: Git/GitHub, Linux (Ubuntu/Debian), WSL2, Docker, VS Code, Anti-gravity IDE, Figma

Achievements

LeetCode: Solved **420+ questions** with a peak contest rating of **1550+**; consistently ranked in the top 35% of biweekly contests.

DroidRun DevSprint : Secured **3rd Position** among 50+ teams at the IIT Patna Hackathon (Feb 2026) for developing the FocusNest AI agent.

Leadership & Experience

InterIIT Tech Meet 14.0

Dec. 2025 – Present

Technical Designer & Web Developer

IIT Patna

- Designed high-fidelity UI/UX in **Figma** and translated prototypes into responsive components for the IITP contingent.

Celesta (Annual Tech-Fest)

Aug. 2025 – Nov. 2025

Flagship Sub-Coordinator

IIT Patna

- Managed technical logistics and rulebooks for flagship robotics events including **Micromouse** and **Line Follower**.

TEDxIITPatna

July 2025 – Jan. 2026

Technical Designer

IIT Patna

- Architected the official website and digital assets using **Figma** for user-flow optimization and brand consistency.